Geocomputation

Spatial Queries and Geometric Operations



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

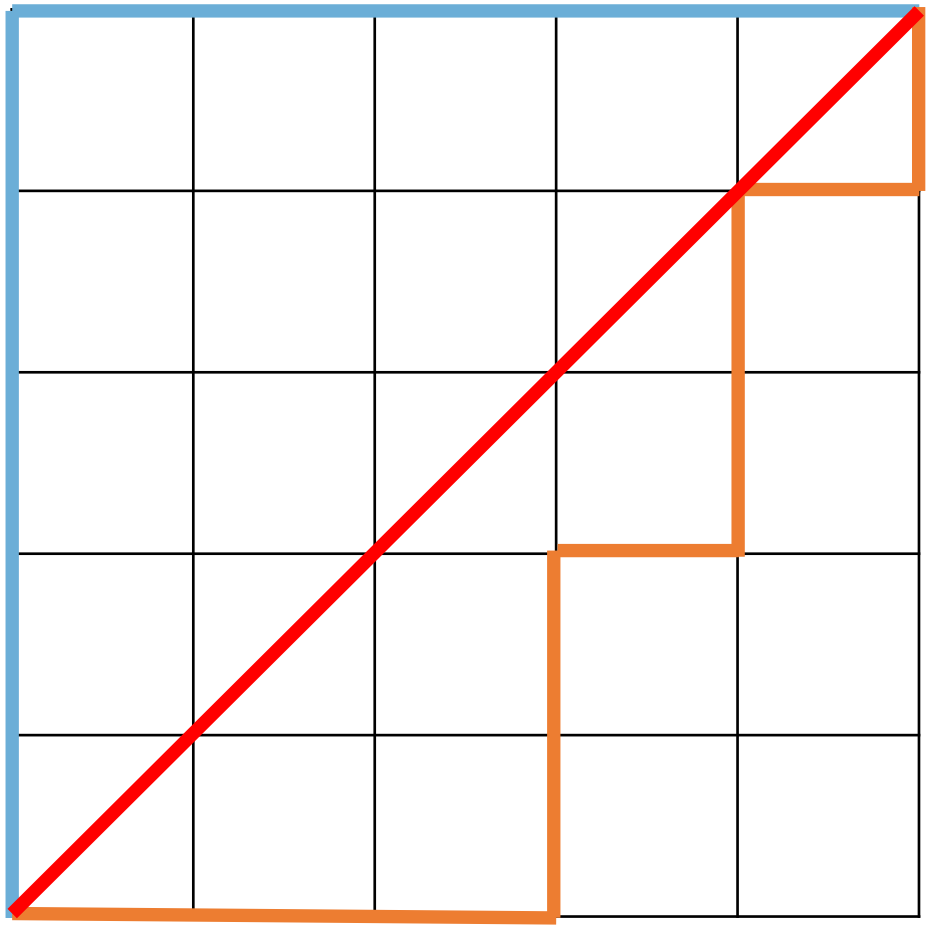
This week

- Spatial properties
- Attribute joins
- Spatial operations

Spatial properties

- Spatial analysis involves performing mathematical operations on spatial properties.
- Key properties include areas of and distances between spatial objects.
- There are diverse ways to conceptualise and work with these properties depending on the analytical context and available data.

Distance: straight line or along a grid

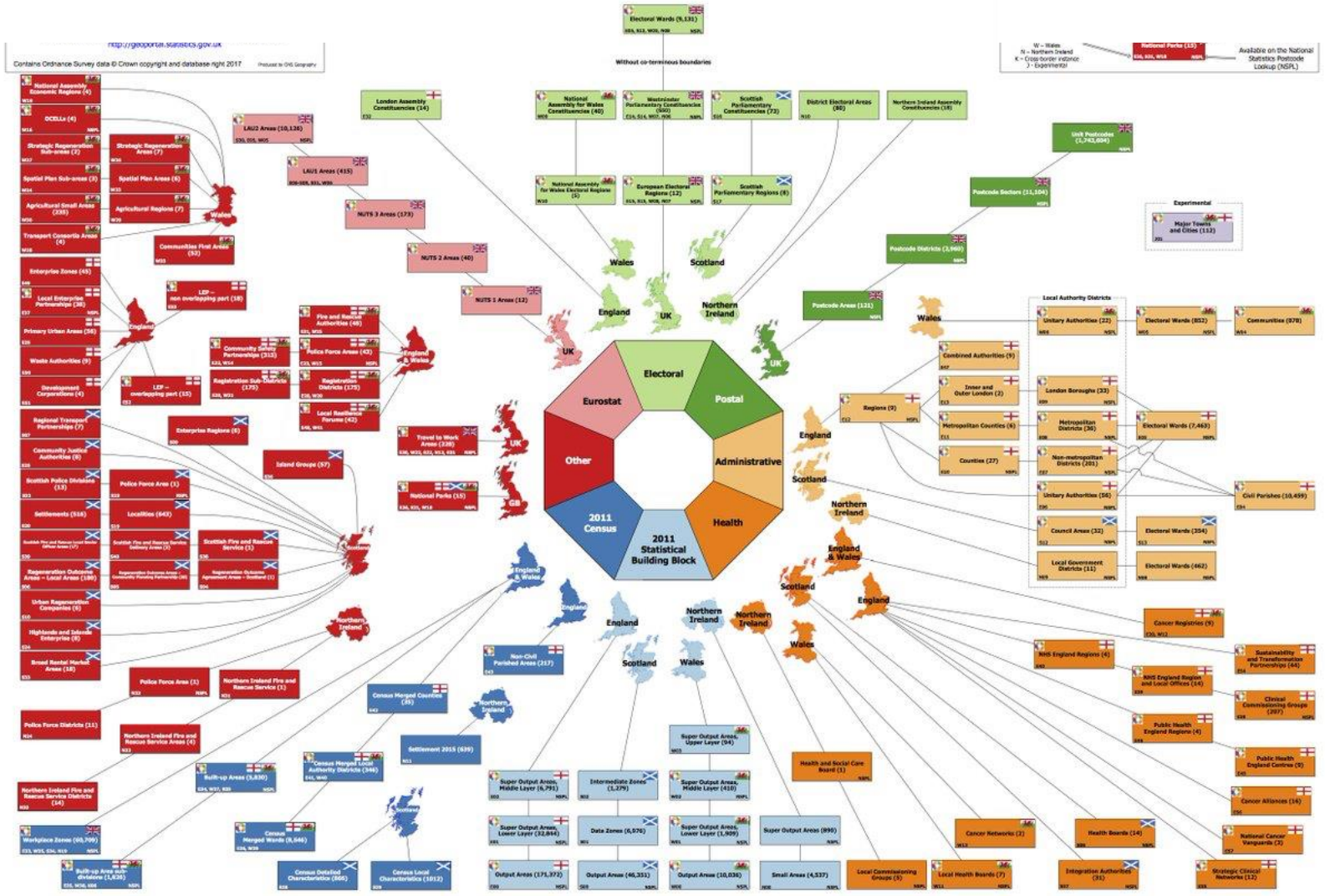


Area

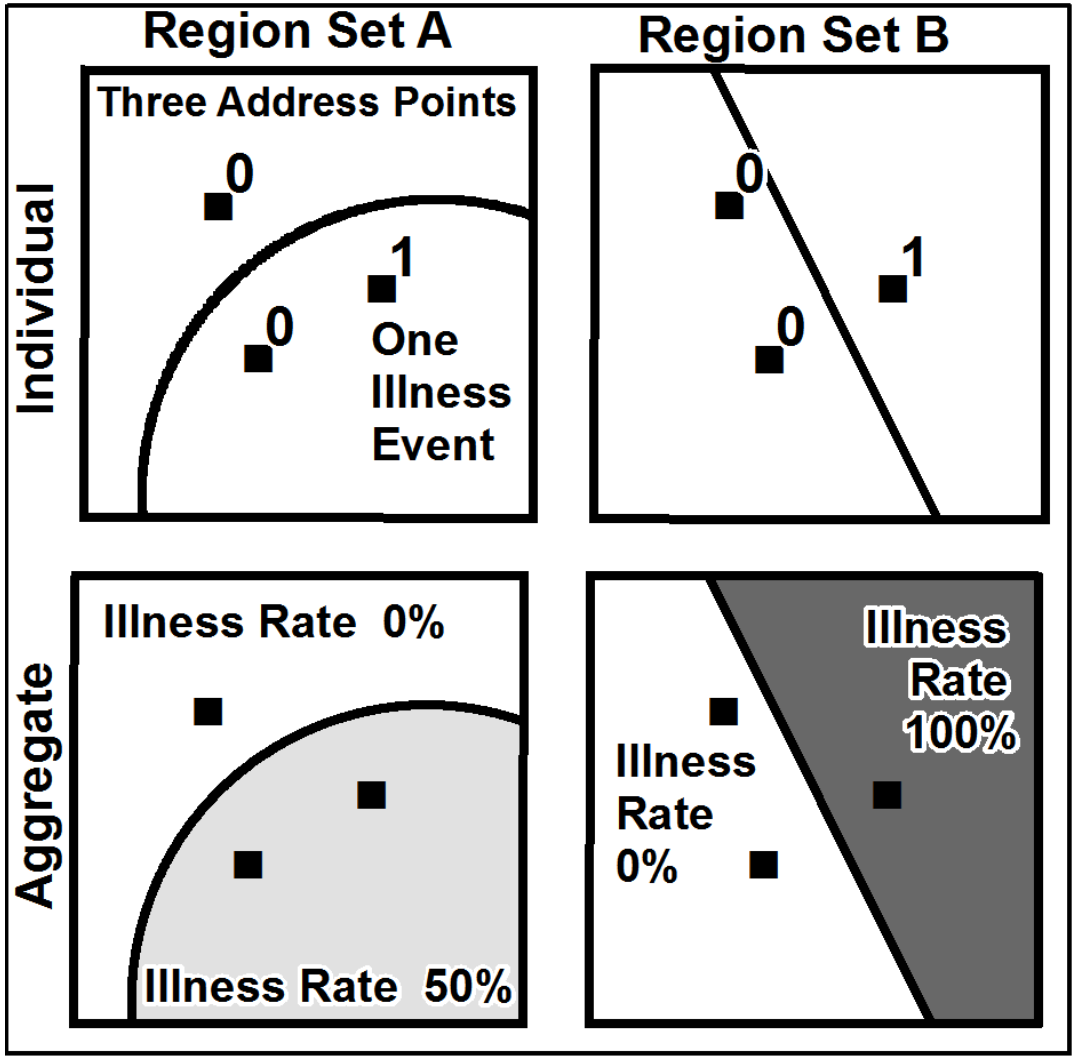




Area: UK statistical geographies



Area: the modifiable areal unit problem



Spatial features

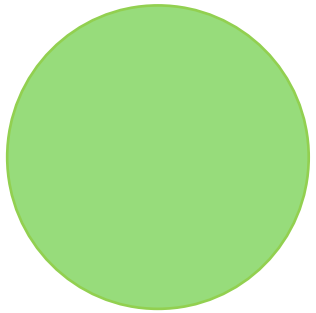
- Features: Objects created in a spatial dataset, representing geographic entities such as points, lines, or polygons.
- Attributes: Characteristics or properties of features, often stored as data in GIS, describing specific details about the feature.

Point vector

Characteristics of a point vector in a GIS data model:

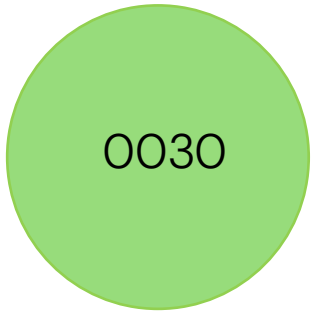
- Single XY location (coordinate).
- Has no area.
- Has no length.
- Geometry consists of a single node or vertex.
- Used for: discrete features or events.

Point vector



Point vector

Point vector



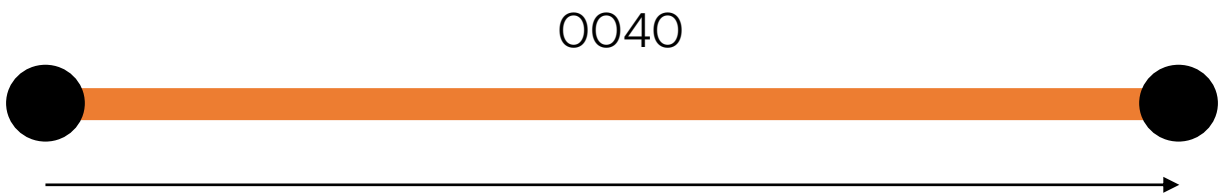
FeatureID	Type	Height
0030	Ent	500

Polyline vector

Characteristics of a polyline vector in a GIS data model:

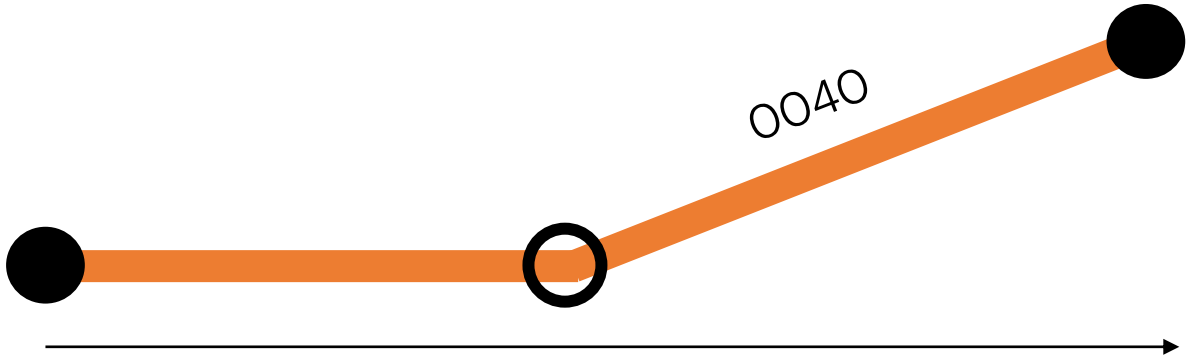
- Series of XY locations (coordinates) that form a line.
- Has no area.
- Has a length.
- Has a direction (importance when it comes to roads, rivers, etc.).
- Can be connected to other polyline vectors to form a network.
- Geometry consists of two nodes (start node and end node) and can have one or more vertices.
- Used for: features without an area but with a length.

Polyline vector



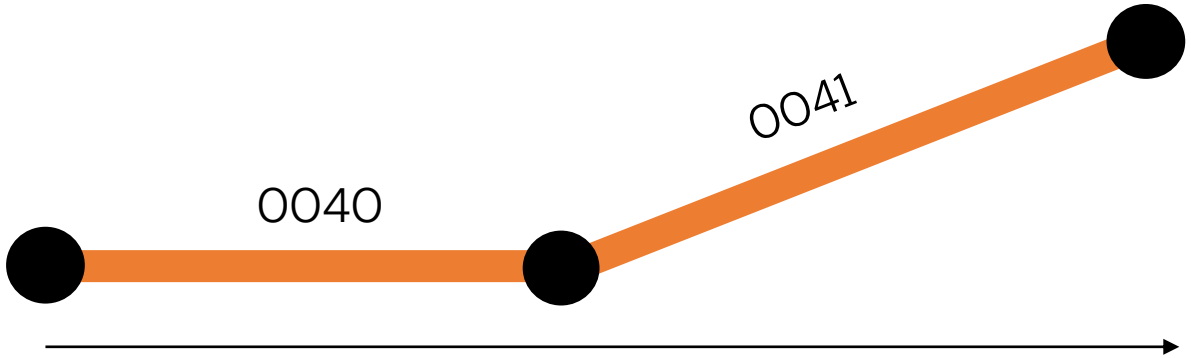
FeatureID	Type	Length (m)
0040	Bicycle lane	1,500

Polyline vector



FeatureID	Type	Length (m)
0040	Bicycle lane	1,650

Polyline vector



FeatureID	Type	Length (m)
0040	Bicycle lane	600
0041	Bicycle lane	1,050

Polygon vector

Characteristics of a polygon vector in a GIS data model:

- Series of XY locations (coordinates) to form an enclosed region.
- Has an area.
- Has a perimeter (the length of its boundary).
- Geometry consists of at least three nodes or vertices whereby the first node or vertex connects with the last one.
- Used for: features with enclosed regions such as buildings and administrative areas.

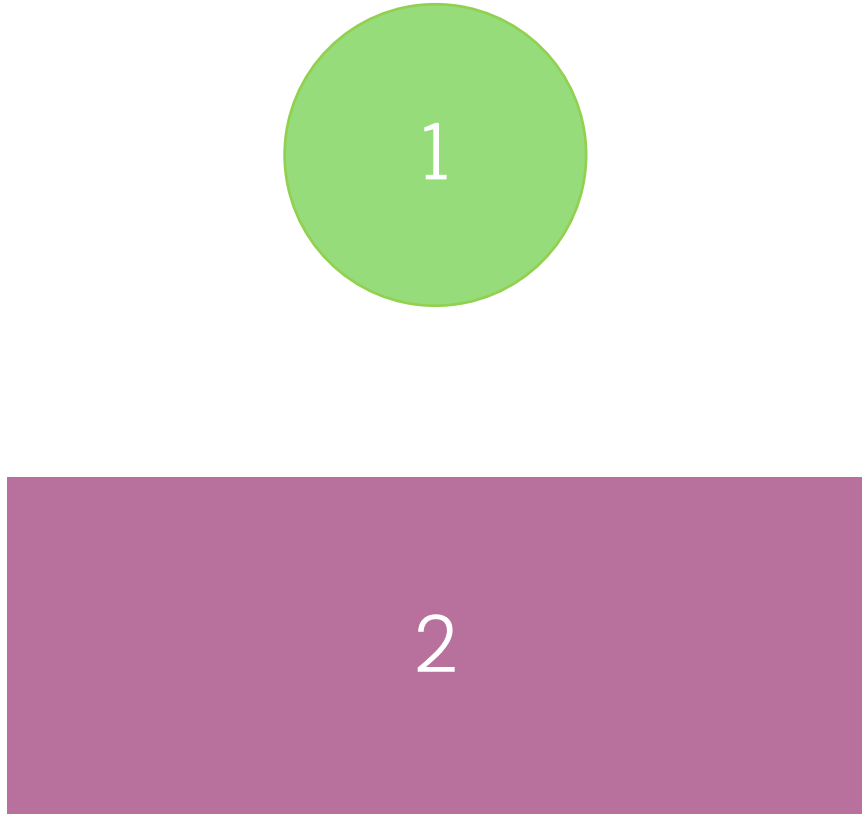
Polygon vector



FeatureID	Type	Area (m²)
0050	University building	2000

Representation

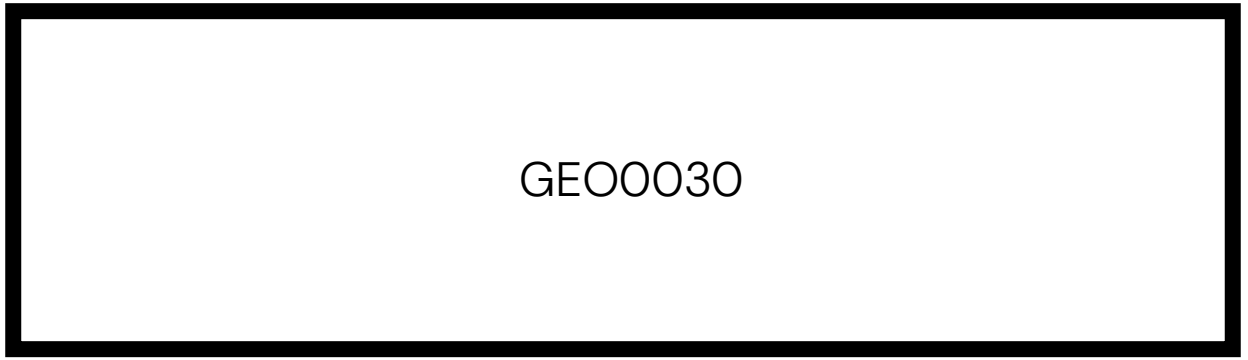
Representing the North-West Wing Building



Attribute join: match rows by a shared ID

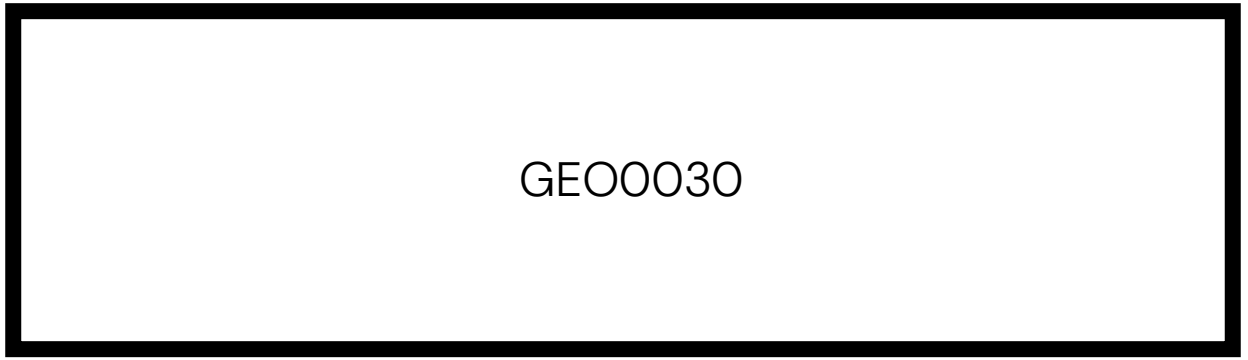
GeoID	Population
GEO0030	540
GEO0031	320

Attribute join: match rows by a shared ID



FeatureID	GeoID
0050	GEO0030

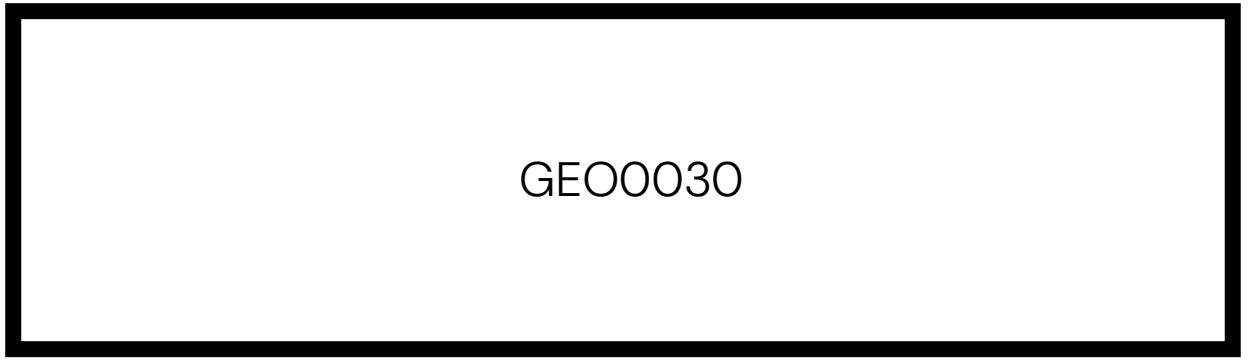
Attribute join: match rows by a shared ID



FeatureID	GeoID
0050	GEO0030

GeoID	Population
GEO0030	540
GEO0031	320

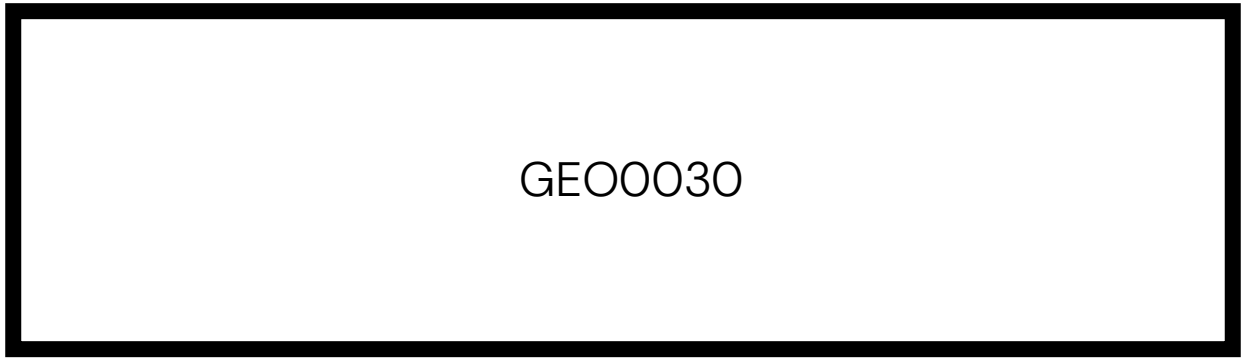
Attribute join: match rows by a shared ID



FeatureID	GeoID
0050	GEO0030

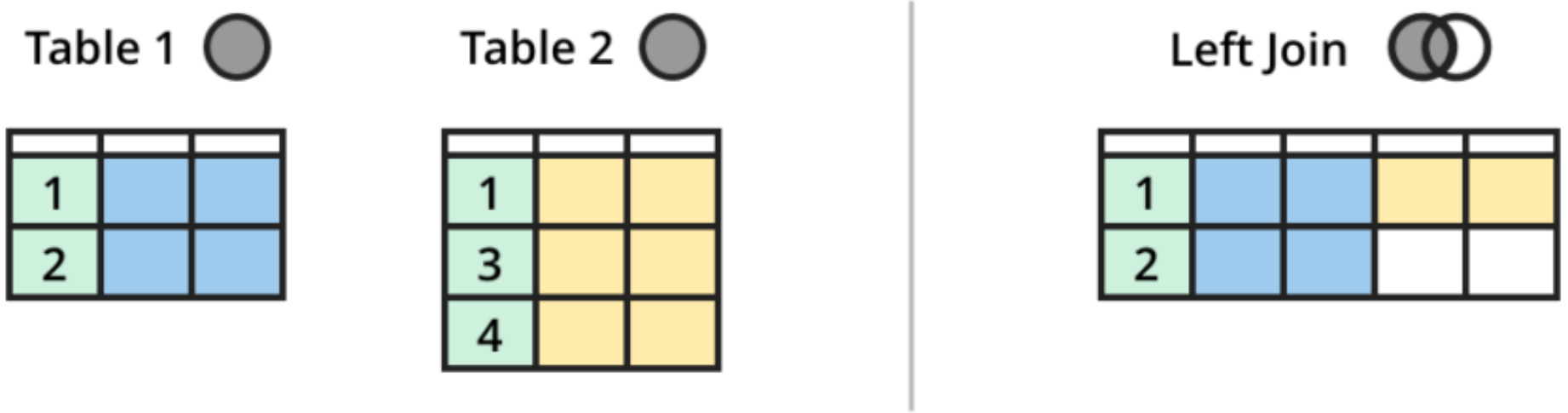
GeoID	Population
GEO0030	540
GEO0031	320

Attribute join: match rows by a shared ID



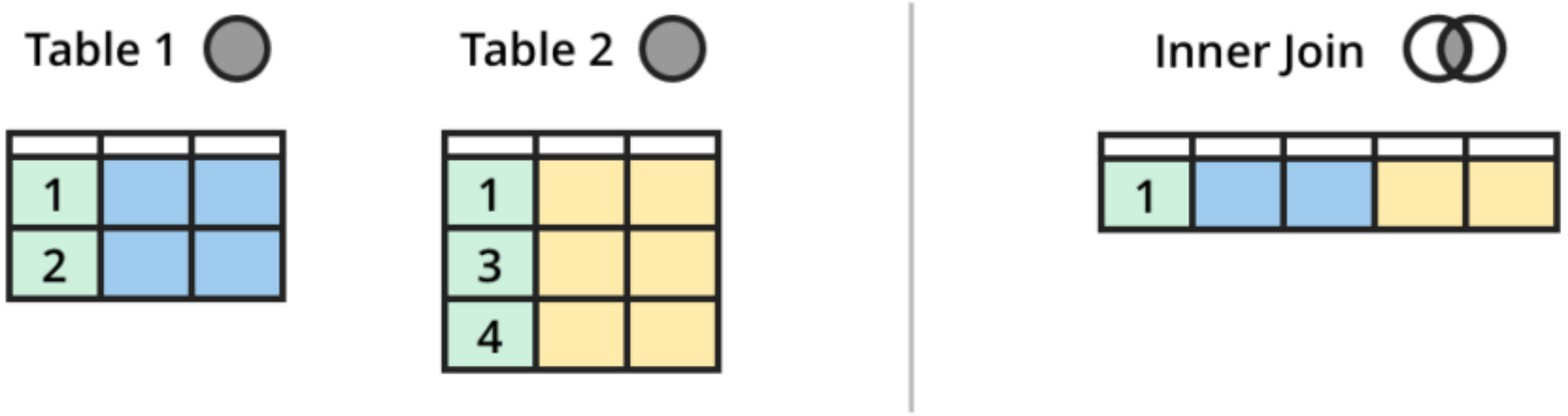
FeatureID	GeoID	Population
0050	GEO0030	540

Left join: keep every row of the first table



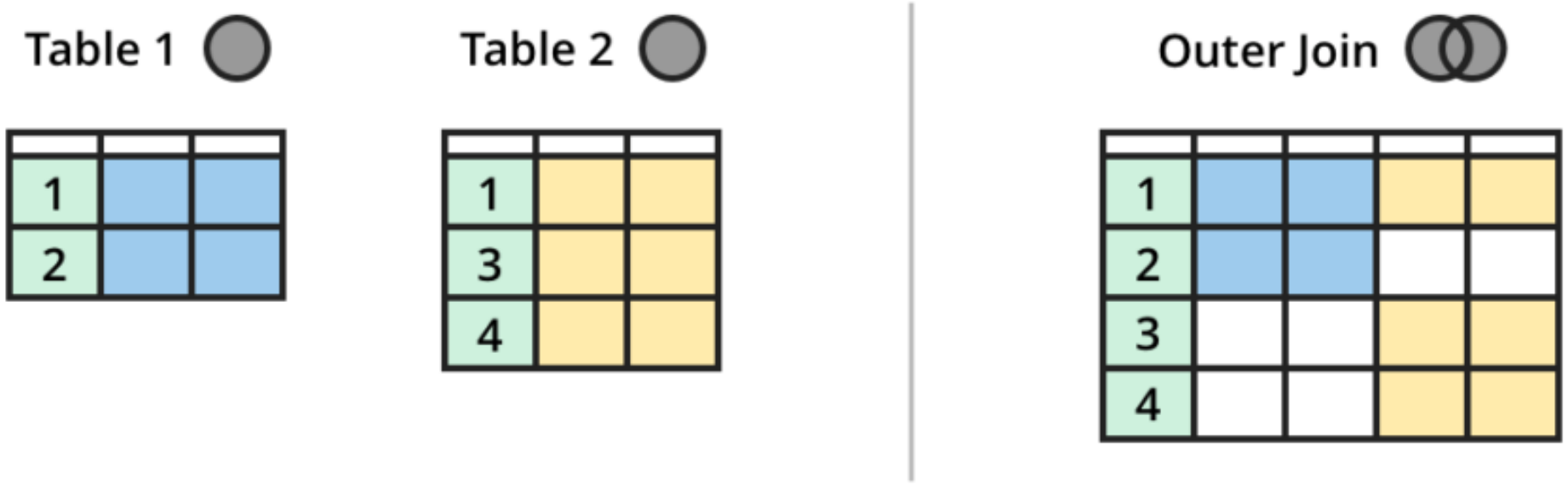
The Data School. 2019. *SQL Join Types Explained Visually*.

Inner join: keep only rows found in both



The Data School. 2019. *SQL Join Types Explained Visually*.

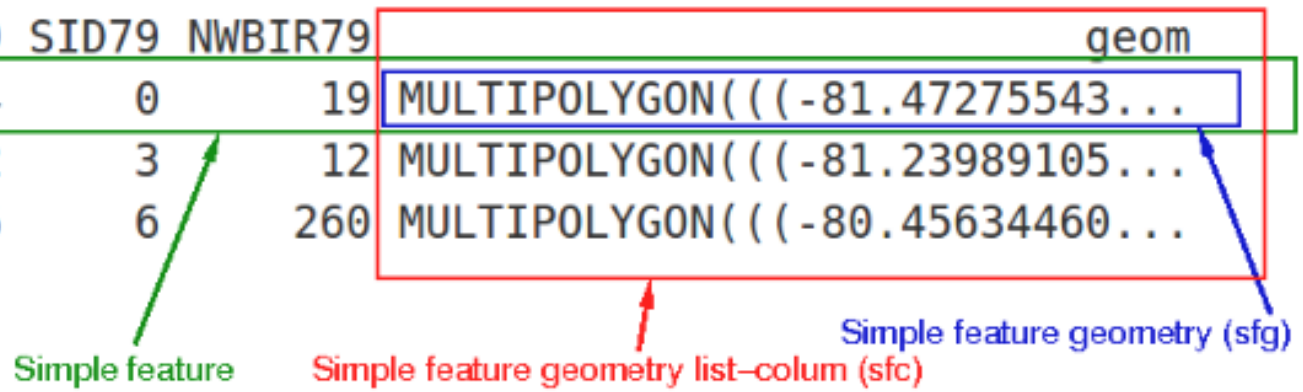
Outer join: keep every row from both tables



The Data School. 2019. *SQL Join Types Explained Visually*.

Spatial data formats

```
## Simple feature collection with 100 features and 6 fields
## geometry type:  MULTIPOLYGON
## dimension:      XY
## bbox:           xmin: -84.32385 ymin: 33.88199 xmax: -75.45698 ymax: 36.58965
## epsg (SRID):    4267
## proj4string:    +proj=longlat +datum=NAD27 +no_defs
## precision:      double (default; no precision model)
## First 3 features:
##   BIR74 SID74 NWBIR74 BIR79 SID79 NWBIR79                              geom
## 1  1091     1       10  1364     0       19 MULTIPOLYGON(((-81.47275543...
## 2   487     0       10   542     3       12 MULTIPOLYGON(((-81.23989105...
## 3  3188     5      208  3616     6      260 MULTIPOLYGON(((-80.45634460...
```



Spatial data formats

- Several commonly used file formats that store spatial data exist.
- Different file formats for vector data and raster data.
- Common vector formats: shapefile, GeoJSON, GeoPackage
- Common raster formats: GeoTIFF, GeoPackage

Shapefiles

- Perhaps the most (in)famous file format.
- Widely used, despite being horribly outdated, especially because of the limitations of the `.dbf` format.
- A shapefile is not a single file, but a collection of files of which at least three are needed for the data to be displayed in GIS software.

Shapefiles

- `.shp` contains the feature geometry..
- `.shx` index file which stores the position of the feature's ID in the `.shp` file.
- `.dbf` stores all attribute information associated with the records..
- `.prj` contains the coordinate system information and projection. *Optional*.
- `.xml` general metadata. *Optional*.
- `.cpg` encoding information. *Optional*.
- `.sbn` optimisation file for spatial queries. *Optional*.

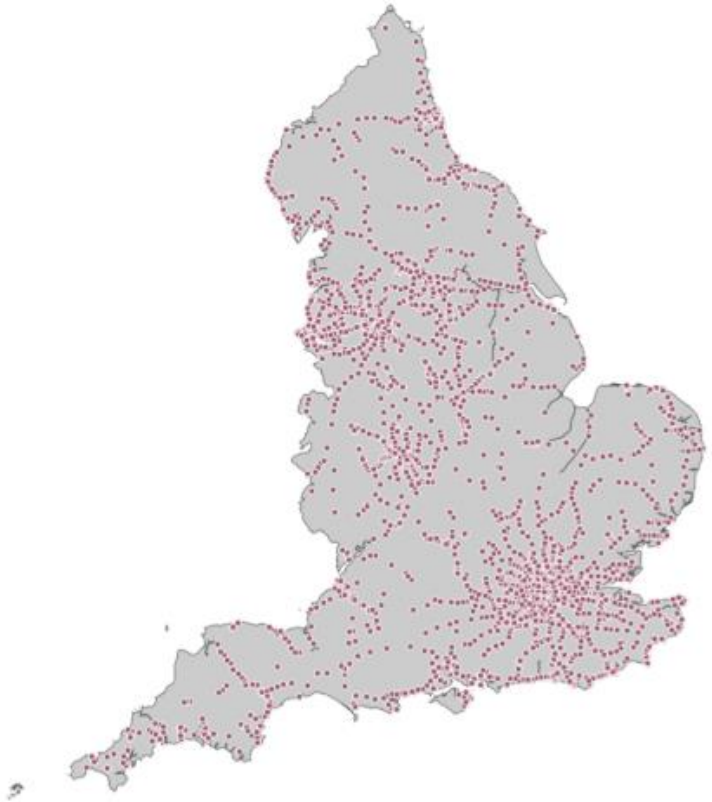
Shapefiles



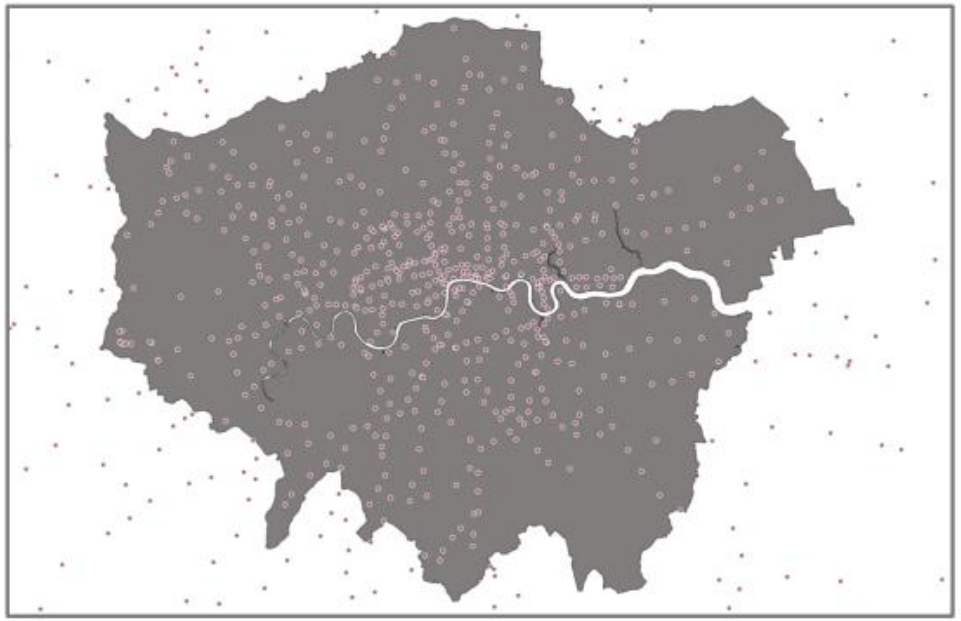
GeoPackage

- A GeoPackage is an open, standards-based, platform-independent, portable, self-describing, compact format for transferring geospatial data.
- It stores spatial data layers in a single file, based upon an SQLite database.
- How to spot in the wild: .gpkg

Spatial operations



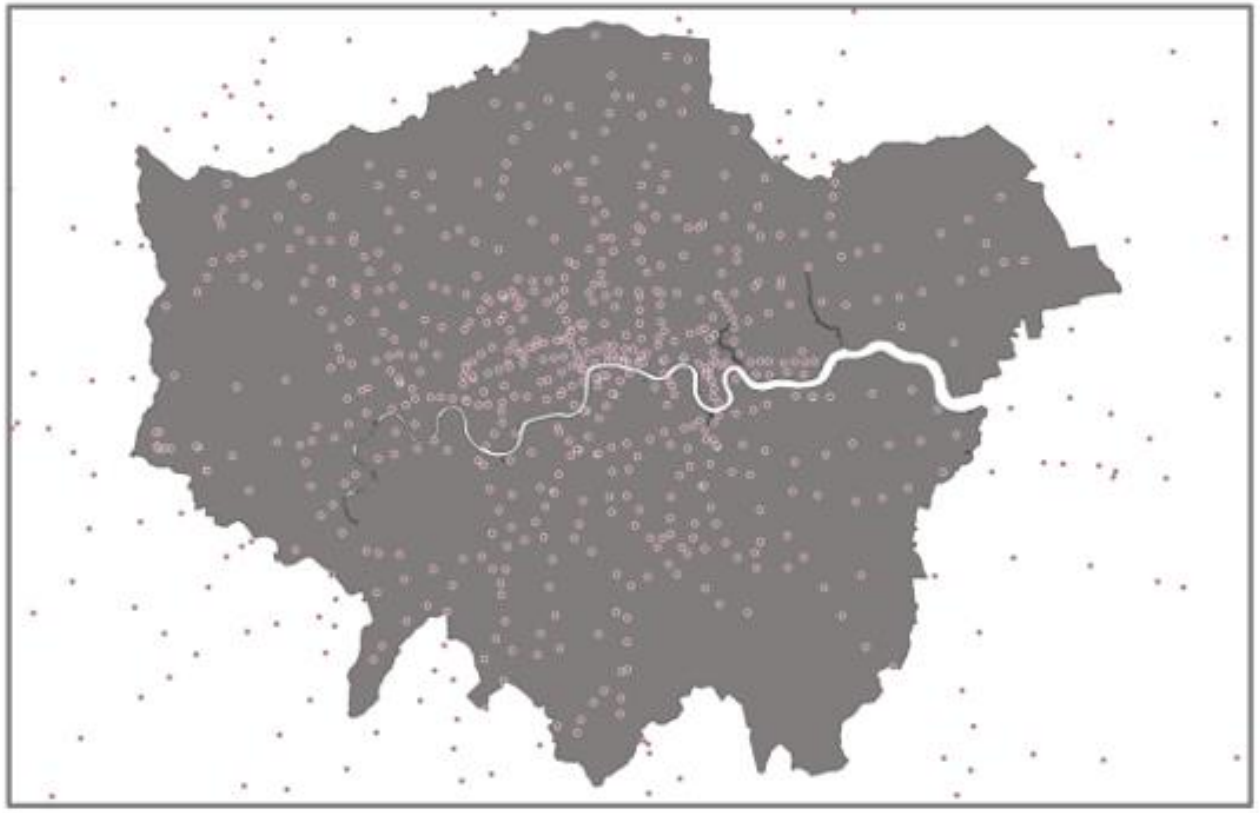
?



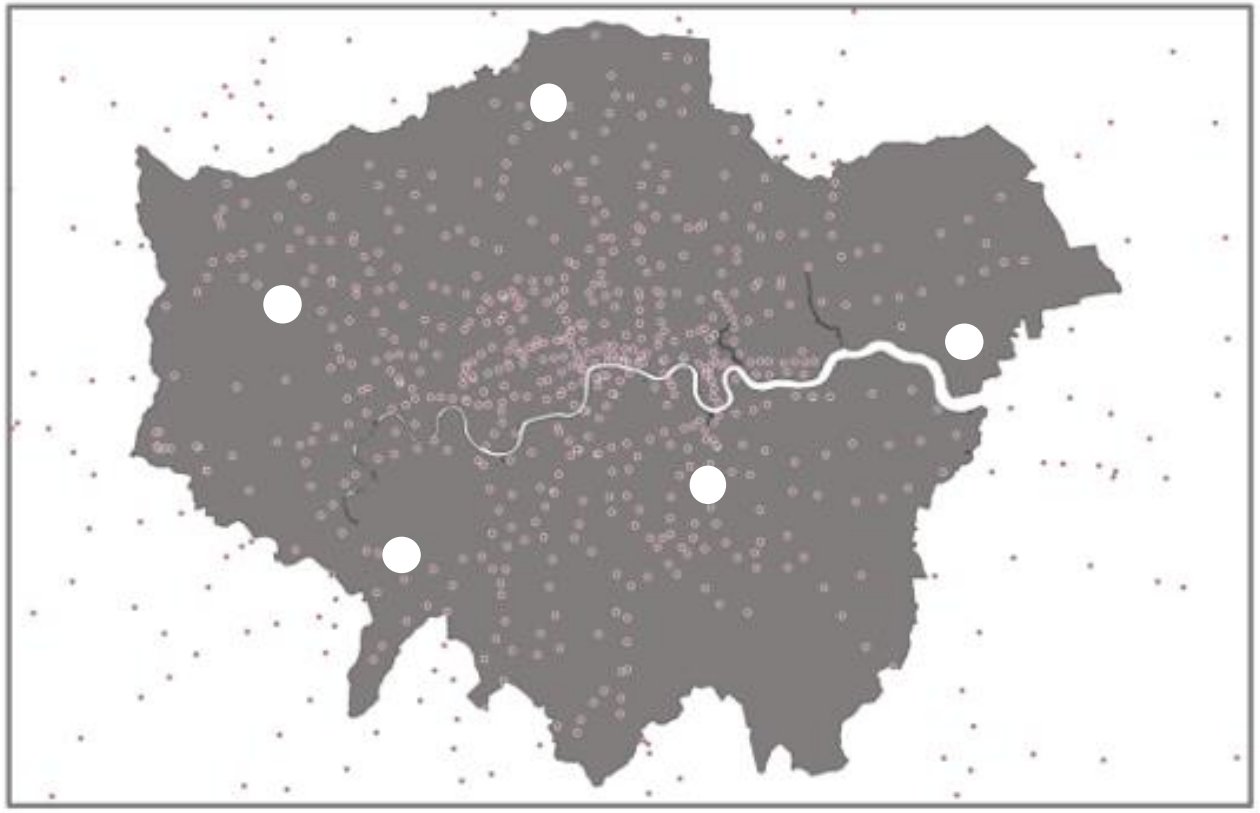
Spatial operations

- Spatial relationships describe how features sit in relation to each other, for example within, touching, overlapping or near, often referred to as topological relationships.
- Spatial operations make use of these spatial relationships between spatial objects to combine, identify, extract information.
- A spatial query selects existing features by where they are; a geometric operation creates new shapes.

Spatial query: select features by location

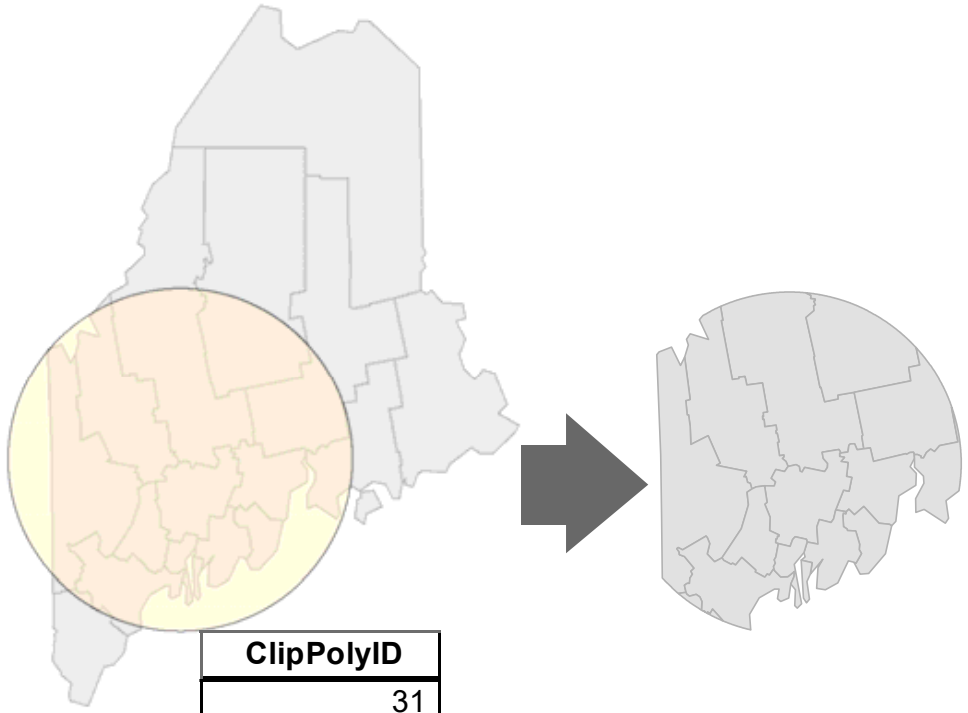


Attribute query: select features by value



Clip: cut one layer to the shape of another

NAME
Androscoggin
Aroostook
Cumberland
Franklin
Hancock
Kennebec
Knox
Lincoln
Oxford
Penobscot
Piscataquis
Sagadahoc
Somerset
Waldo
Washington
York



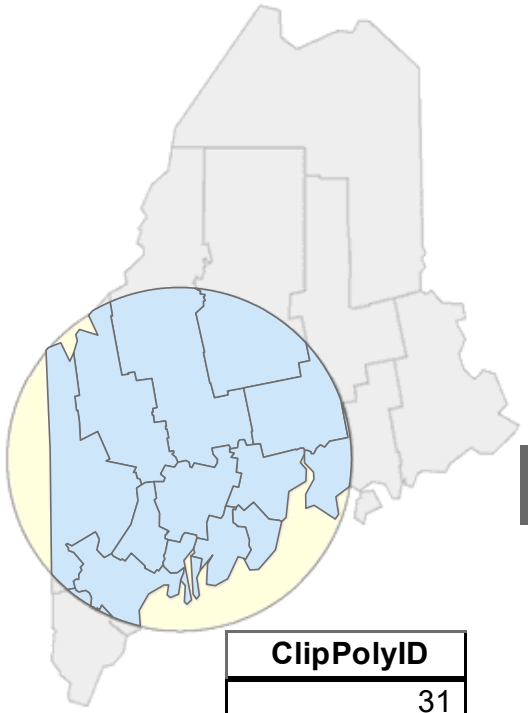
ClipPolyID
31

NAME
Androscoggin
Cumberland
Franklin
Hancock
Kennebec
Knox
Lincoln
Oxford
Penobscot
Piscataquis
Sagadahoc
Somerset
Waldo
York

Gimond, M. 2025. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/index.html

Intersect: keep only where two layers overlap

NAME
Androscoggin
Aroostook
Cumberland
Franklin
Hancock
Kennebec
Knox
Lincoln
Oxford
Penobscot
Piscataquis
Sagadahoc
Somerset
Waldo
Washington
York



ClipPolyID
31

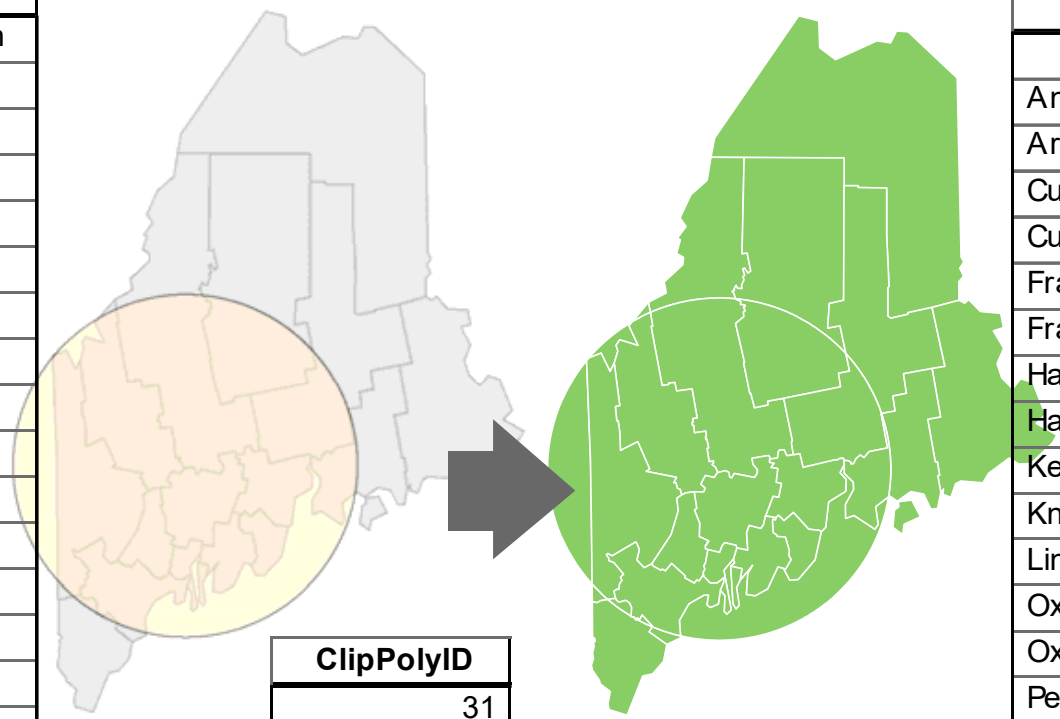


NAME	ClipPolyID
Androscoggin	31
Cumberland	31
Franklin	31
Hancock	31
Kennebec	31
Knox	31
Lincoln	31
Oxford	31
Penobscot	31
Piscataquis	31
Sagadahoc	31
Somerset	31
Waldo	31
York	31

Gimond, M. 2025. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/index.html

Union: combine two layers, keep every area

NAME
Androscoggin
Aroostook
Cumberland
Franklin
Hancock
Kennebec
Knox
Lincoln
Oxford
Penobscot
Piscataquis
Sagadahoc
Somerset
Waldo
Washington
York

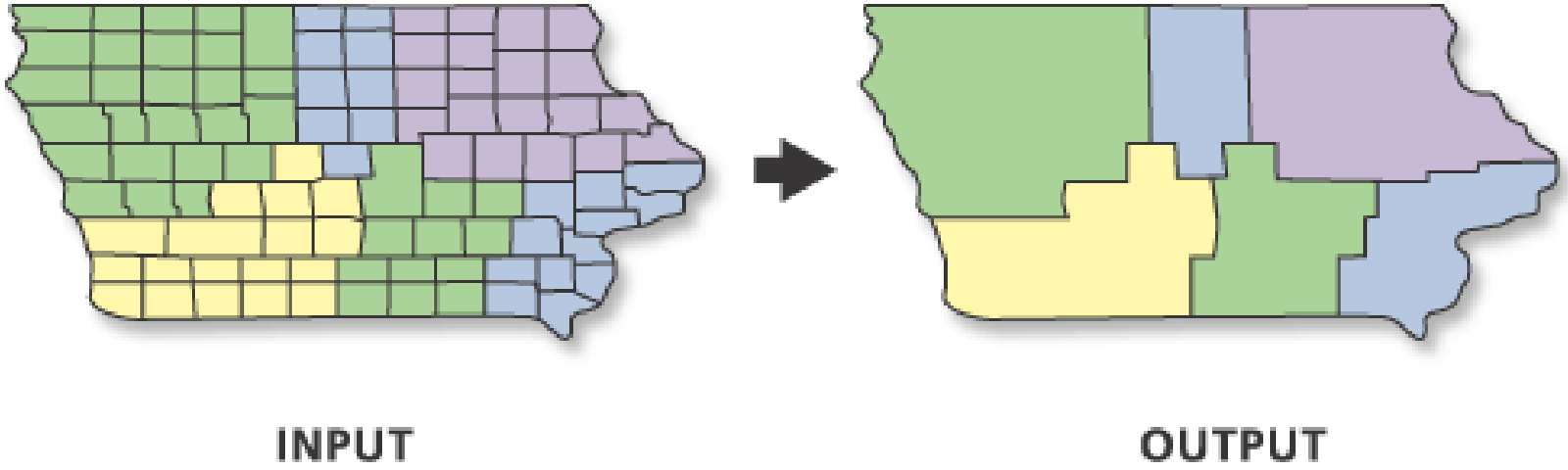


ClipPolyID
31

NAME	ClipPolyID
	31
Androscoggin	31
Aroostook	0
Cumberland	0
Cumberland	31
Franklin	0
Franklin	31
Hancock	0
Hancock	31
Kennebec	31
Knox	31
Lincoln	31
Oxford	0
Oxford	31
Penobscot	0

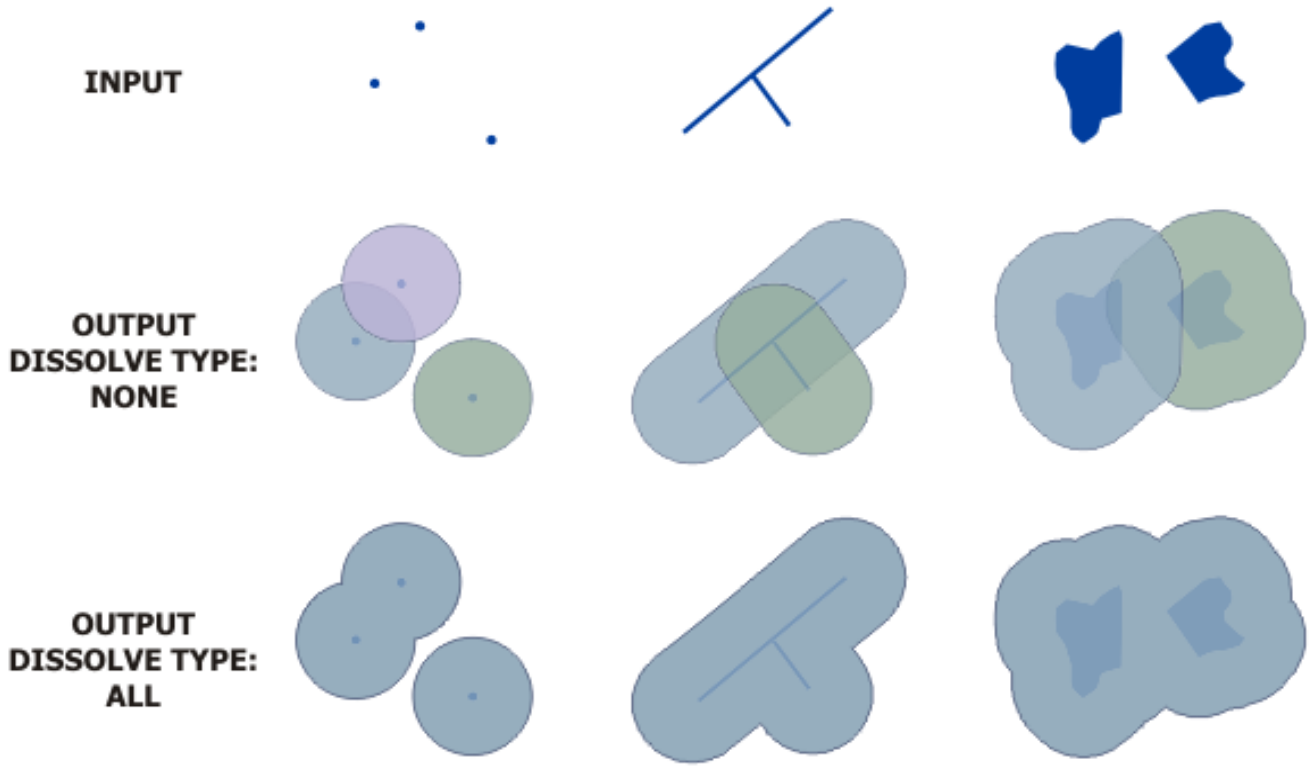
Gimond, M. 2025. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/index.html

Dissolve: merge features into one



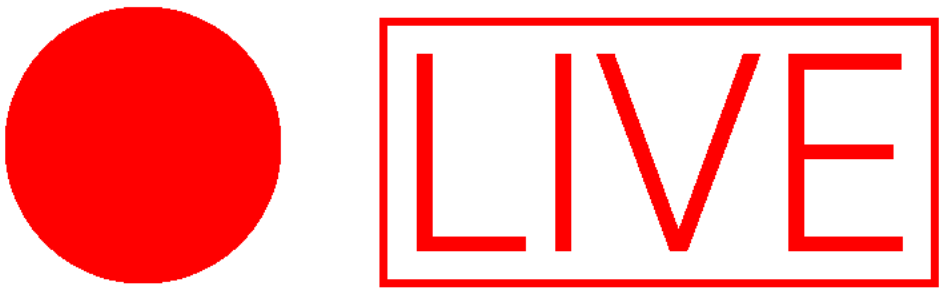
ESRI. 2025. Dissolve. [online]
https://pro.arcgis.com/en/pro-app/latest/tool-reference/data-management/dissolve.htm

Buffer: draw a zone of fixed distance



ESRI. 2025. Buffer (Analysis). [online]
https://pro.arcgis.com/en/pro-app/latest/tool-reference/analysis/buffer.htm

RStudio



Conclusion

- The core of spatial analysis comes down to conducting spatial queries and executing geometric operations.
- Spatial analysis therefore relies on the spatial properties of an object as well as on the spatial relationships both *within* and *between* spatial objects.

Questions

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